

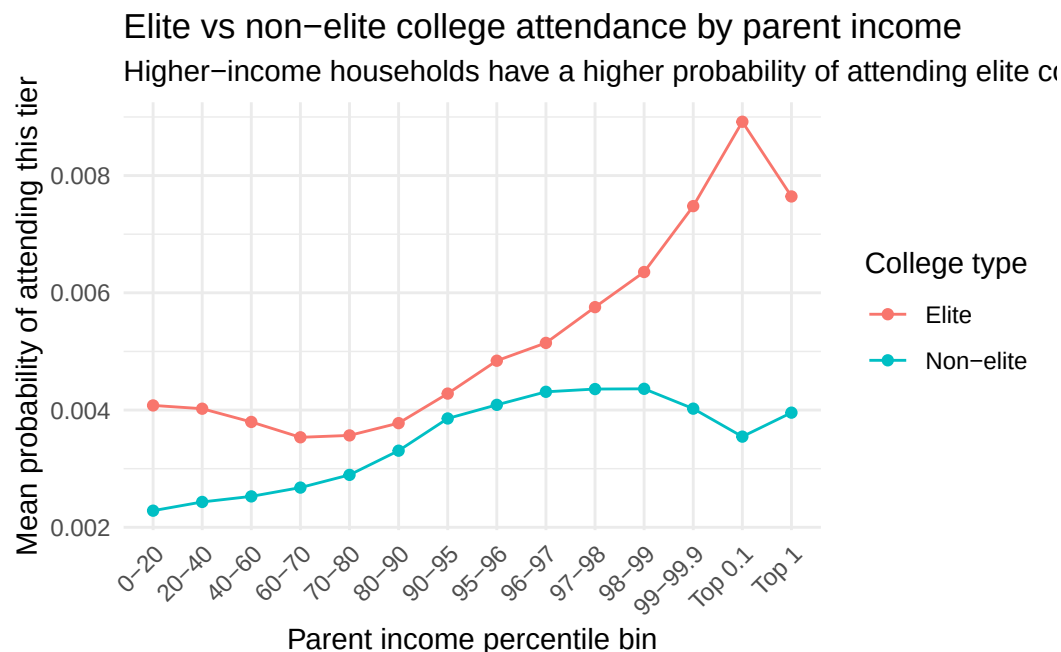
Link between Elite College Admission and high household income

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Introduction paragraph

In this project, I focus on how family income is related to who gets into elite colleges. Using a college admissions dataset with detailed parent income bins and college “tier” labels, I look at how the probability of attending an elite college changes as household income goes up. The SAT data I include is more of a supporting piece: it shows that higher-income students also tend to have higher test scores, which can help explain why they show up more often at elite schools. Overall, my goal is to use graphs and a hypothesis test to see whether students from high-income households are clearly more likely to attend elite colleges than students from lower-income households.



Hypothesis test for correlation

Here I test whether there is a statistically significant (positive) correlation between parent income and the chance of attending an **elite** college.

- **Null hypothesis (H₀):** $\rho = 0$ (no linear correlation between income bin and elite attendance).
- **Alternative hypothesis (H_a):** $\rho > 0$ (higher income bins are associated with higher elite attendance).

Pearson's product-moment correlation

```
data: income_order and attend
t = 10.714, df = 1496, p-value < 2.2e-16
alternative hypothesis: true correlation is greater than 0
95 percent confidence interval:
 0.2270053 1.0000000
sample estimates:
      cor
0.2669443
```

Estimated correlation (rho): 0.267

p-value: 3.7e-26

Data wrangling

To see if there another link we used an SAT dataset involving scores and median family income. Here is how we wrangled the data.

	income_quintile	income_range	n_test_takers	pct_test_takers
1	Lowest Quintile	\$0 - \$59,669	173521	0.09
2	2nd Lowest Quintile	\$59,670 - \$76,797	196975	0.10
3	Middle Quintile	\$76,798 - \$95,223	233173	0.12
4	2nd Highest Quintile	\$95,224 - \$124,913	319127	0.16
5	Highest Quintile	>= \$124,914	494083	0.25
6	Unknown	Unknown	588086	0.29

	mean_total	mean_erw	mean_math	pct_bench_both	pct_bench_erw	pct_bench_math
1	897	460	438	0.16	0.42	0.17
2	951	488	463	0.24	0.54	0.25
3	994	510	485	0.31	0.62	0.33
4	1049	536	513	0.42	0.72	0.43
5	1161	589	572	0.63	0.86	0.64
6	985	490	495	0.33	0.52	0.37

	pct_bench_none
1	0.56
2	0.45
3	0.37
4	0.27

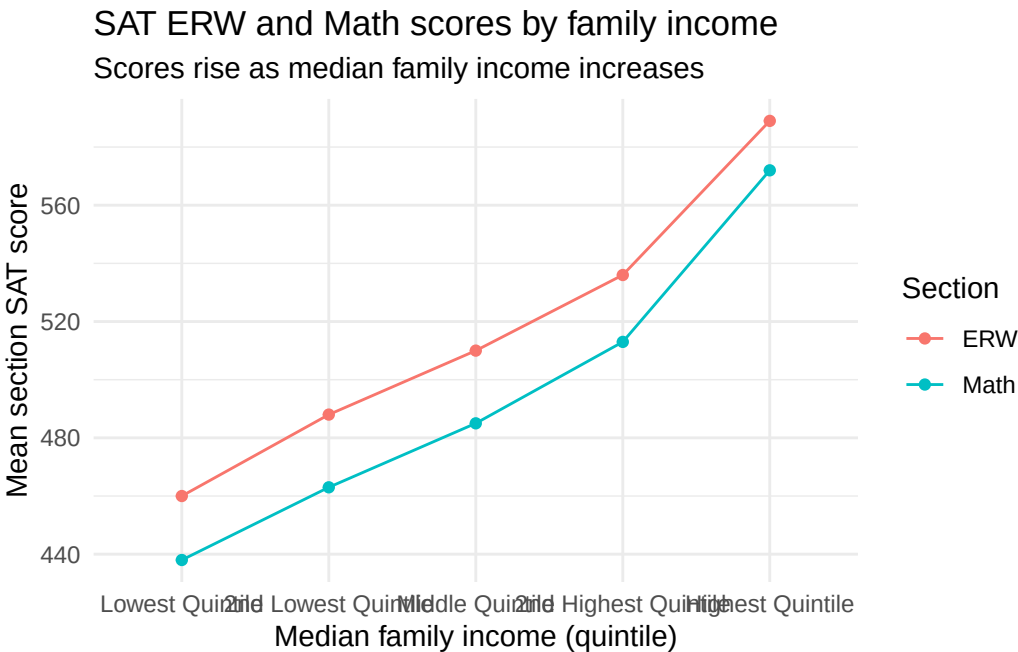
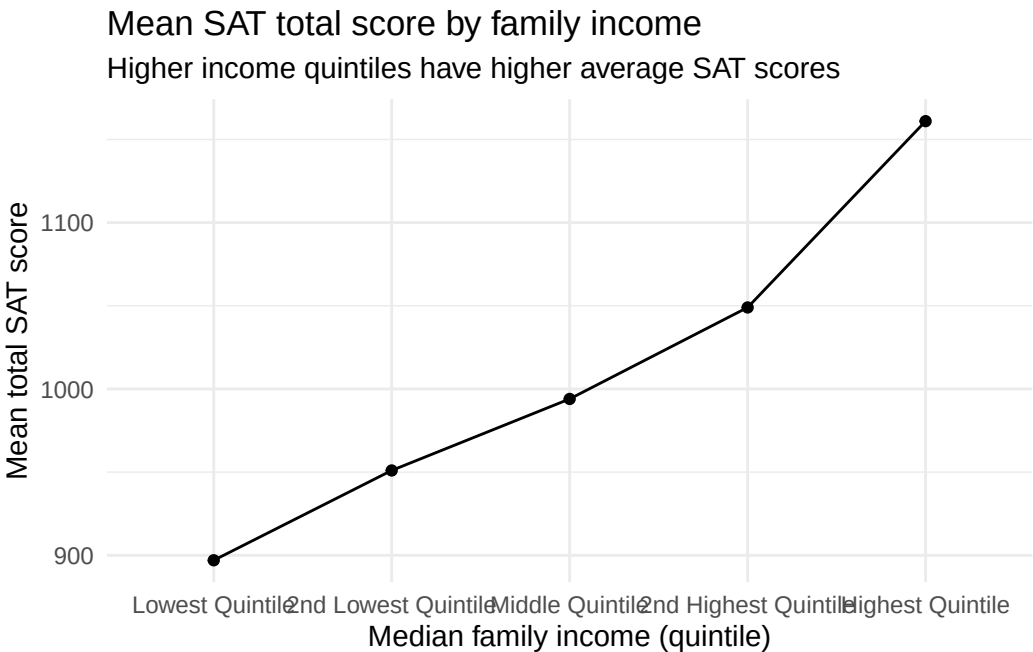
5 0.13
6 0.44

	income_quintile	income_range	score_type	mean_score
1	Lowest Quintile	\$0 - \$59,669	Total	897
2	2nd Lowest Quintile	\$59,670 - \$76,797	Total	951
3	Middle Quintile	\$76,798 - \$95,223	Total	994
4	2nd Highest Quintile	\$95,224 - \$124,913	Total	1049
5	Highest Quintile	>= \$124,914	Total	1161
6	Unknown	Unknown	Total	985
7	Lowest Quintile	\$0 - \$59,669	ERW	460
8	2nd Lowest Quintile	\$59,670 - \$76,797	ERW	488
9	Middle Quintile	\$76,798 - \$95,223	ERW	510
10	2nd Highest Quintile	\$95,224 - \$124,913	ERW	536
11	Highest Quintile	>= \$124,914	ERW	589
12	Unknown	Unknown	ERW	490
13	Lowest Quintile	\$0 - \$59,669	Math	438
14	2nd Lowest Quintile	\$59,670 - \$76,797	Math	463
15	Middle Quintile	\$76,798 - \$95,223	Math	485
16	2nd Highest Quintile	\$95,224 - \$124,913	Math	513
17	Highest Quintile	>= \$124,914	Math	572
18	Unknown	Unknown	Math	495

	income_quintile	income_range	benchmark_type	pct_met
1	Lowest Quintile	\$0 - \$59,669	Both	0.16
2	2nd Lowest Quintile	\$59,670 - \$76,797	Both	0.24
3	Middle Quintile	\$76,798 - \$95,223	Both	0.31
4	2nd Highest Quintile	\$95,224 - \$124,913	Both	0.42
5	Highest Quintile	>= \$124,914	Both	0.63
6	Unknown	Unknown	Both	0.33
7	Lowest Quintile	\$0 - \$59,669	ERW	0.42
8	2nd Lowest Quintile	\$59,670 - \$76,797	ERW	0.54
9	Middle Quintile	\$76,798 - \$95,223	ERW	0.62
10	2nd Highest Quintile	\$95,224 - \$124,913	ERW	0.72
11	Highest Quintile	>= \$124,914	ERW	0.86
12	Unknown	Unknown	ERW	0.52
13	Lowest Quintile	\$0 - \$59,669	Math	0.17
14	2nd Lowest Quintile	\$59,670 - \$76,797	Math	0.25
15	Middle Quintile	\$76,798 - \$95,223	Math	0.33
16	2nd Highest Quintile	\$95,224 - \$124,913	Math	0.43
17	Highest Quintile	>= \$124,914	Math	0.64
18	Unknown	Unknown	Math	0.37
19	Lowest Quintile	\$0 - \$59,669	None	0.56
20	2nd Lowest Quintile	\$59,670 - \$76,797	None	0.45
21	Middle Quintile	\$76,798 - \$95,223	None	0.37
22	2nd Highest Quintile	\$95,224 - \$124,913	None	0.27
23	Highest Quintile	>= \$124,914	None	0.13
24	Unknown	Unknown	None	0.44

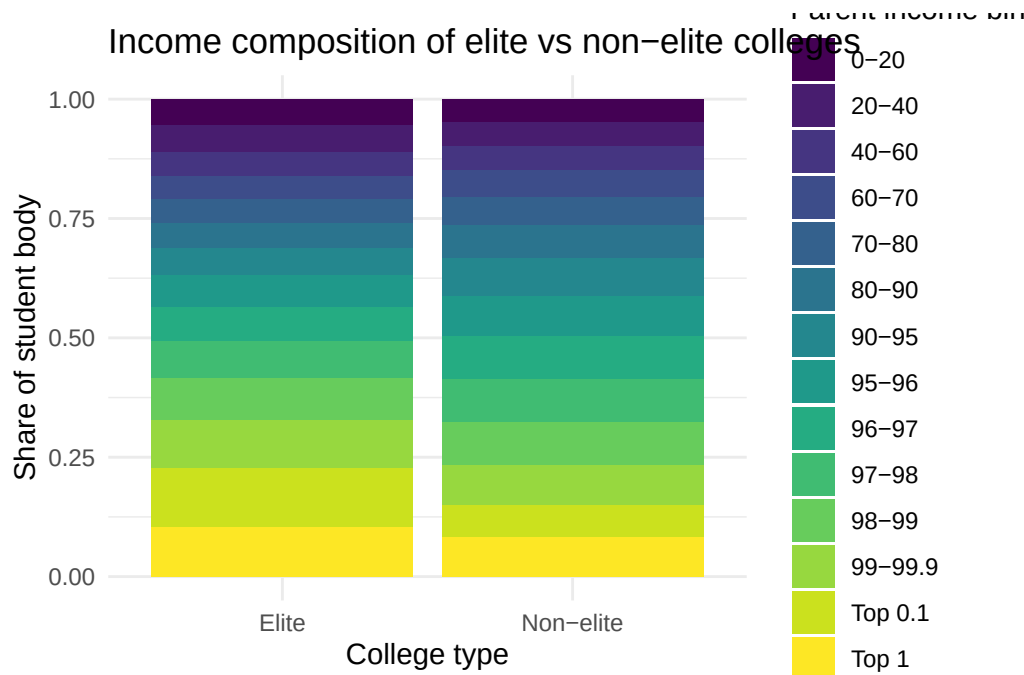
SAT Scores vs Median household income plot

After looking at elite college attendance, we also want to see whether test performance itself is related to family income. This SAT table shows that students from higher-income quintiles have higher mean SAT scores and are more likely to meet both ERW and Math benchmarks, which helps explain why they're overrepresented at elite colleges.

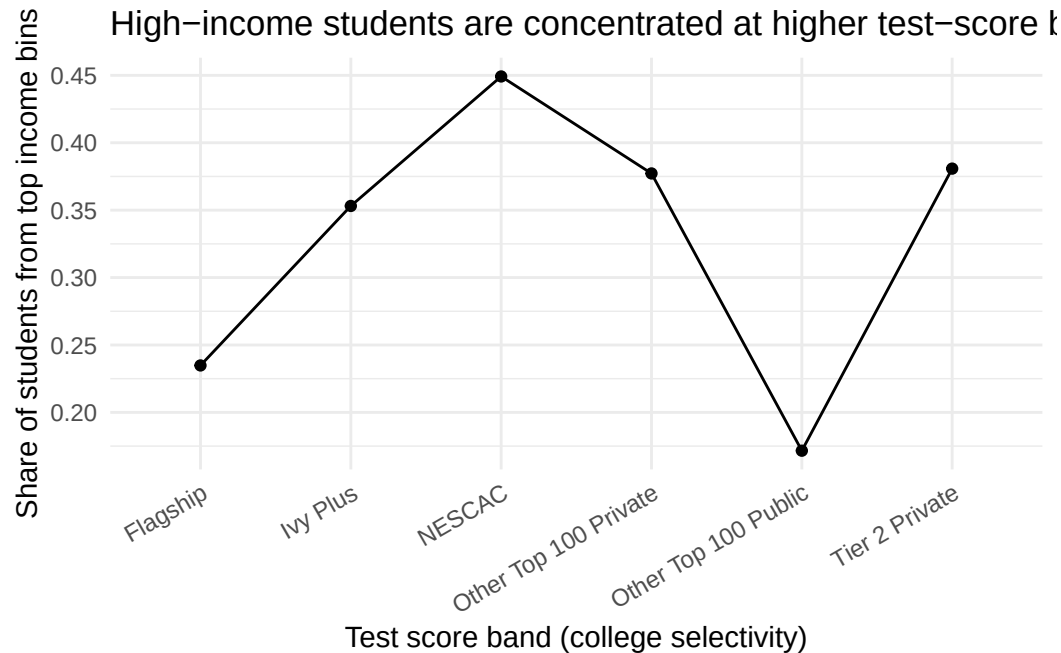


Additional Graphs supporting the link between household income and Elite college admission

These three graphs give more evidence that wealth and elite college access are tightly connected. The stacked bar chart shows that elite colleges enroll a much larger share of students from the very highest income bins than non-elite colleges. The heatmap shows that, within tiers, attendance probabilities are highest for rich students at the most selective tiers. Finally, the high-income-share vs test-score-band graph connects this back to performance: colleges with stronger test-score profiles have a bigger share of high-income students, reinforcing the idea that family income, test scores, and elite admissions are all linked.



```
# A tibble: 6 x 3
# Groups:   elite_flag [2]
  elite_flag bin_group share
  <chr>      <chr>    <dbl>
1 Elite     High      0.529
2 Elite     Low       0.207
3 Elite     Middle    0.264
4 Non-elite High      0.440
5 Non-elite Low       0.203
6 Non-elite Middle    0.357
```



##Conclusion Overall, my results show a clear and consistent link between higher household income and elite college attendance. The line graph and hypothesis test both point in the same direction: as parent income increases, the probability of attending an elite college rises, and this positive correlation is statistically significant, not just random noise. The composition table also shows that elite colleges are made up of a much larger share of students from the highest income bins and a smaller share from the lowest bins compared to non-elite schools. The SAT data is only supplementary, but it supports this pattern by showing that higher-income students tend to have higher test scores, which likely helps them access these selective institutions. Taken together, the evidence suggests that elite college admissions are strongly tilted toward students from wealthier families.